



OPEN Wettability-Driven void suppression and enhanced mechanical performance in Si_3N_4 -Reinforced bamboo Fiber epoxy composites using COCOSO and ANN modeling

M. Saravana Kumar^{1,2}✉, Che-Hua Yang¹, Khaja Moiduddin³, V. Kavimani^{4,5}, Adeolu A. Adediran^{6,7}✉ & Syed Hammad Mian³

The incorporation of ceramic particles as fillers and natural fibers as reinforcements in polymer composites has gained significant attention in recent years. However, defects such as particle agglomeration, poor wettability, non-homogeneous distribution, and void formation often arise due to unoptimized process parameters. This study aims to enhance wettability between fillers, reinforcement, and the matrix while improving uniform dispersion by optimizing filler percentage, stirring time, and stirring speed. To achieve this, Silicon Nitride (Si_3N_4) fillers were dispersed in bamboo fiber-reinforced epoxy composites using an L_{27} orthogonal array, with process parameters varied across Si_3N_4 filler content (3%, 6%, and 9%), stirring time (6, 12, and 18 min), and stirring speed (250, 350, and 450 rpm). The response parameters such as void formation, contact angle, and tensile strength were analyzed, and the Combined Compromise Solution (COCOSO) method, integrated with an Artificial Neural Network (ANN), was used to determine the optimal parameter settings and compare experimental results with predictions. The findings reveal that using 6% Si_3N_4 , a stirring time of 12 min, and a stirring speed of 350 rpm improves wettability by 25.1% while reducing void formation by 79.3%. These results highlight the importance of optimizing wettability through precise control of filler content and processing parameters to achieve high-performance composites with minimal defects and enhanced mechanical properties.

Keywords Bamboo, Particles, Silicon nitride, Voids, Epoxy composites

With increasing ecological and environmental awareness, significant efforts have been devoted to developing sustainable, eco-friendly, and innovative materials for various applications¹. Bamboo fiber-reinforced epoxy composites, a type of environmentally friendly biocomposite, offer benefits such as moderate strength, lightweight properties, water resistance, high durability, and recyclability, making them a subject of growing research interest². However, using these materials as reinforcements in epoxy composites often fails to deliver the necessary structural integrity. To address this challenge and boost overall performance, this study introduces silicon nitride (Si_3N_4) particles to enhance the mechanical strength of the composites. Furthermore, critical

¹Graduate Institute of Manufacturing Technology, National Taipei University of Technology, Taipei 10608, Taiwan.

² Department of Mechanical Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences (SIMATS), Chennai 602105, Tamil Nadu, India. ³Advanced Manufacturing Institute, King Saud University, P.O. Box 800, Riyadh 11421, Saudi Arabia. ⁴Department of Mechanical Engineering, Karpagam Academy of Higher Education, Coimbatore 641021, Tamil Nadu, India. ⁵Centre for material science, Karpagam Academy of Higher Education, Coimbatore 641021, Tamil Nadu, India. ⁶Materials Design and Structural Integrity Group, Department of Materials and Metallurgical Engineering, Federal University Oye-Ekiti, Ikole-Ekiti, Ekiti State, Nigeria. ⁷Department of Mechanical Engineering Science, University of Johannesburg, Johannesburg, South Africa. ✉email: saravana312@gmail.com; dladesoji@yahoo.com

process parameters, such as filler loading, stirring speed, and stirring duration, are systematically examined for their effects on void formation, contact angle, and tensile properties, with the goal of identifying the optimal conditions for producing high-quality, defect-free composites.

The mechanical properties and morphology of HDPE/bamboo flour (BF) composites modified with PE-g-MA and EPR-g-MA at different concentrations were examined by Liu et al. Experimental results indicated that PE-g-MA increased strength and stiffness, while sEPR-g-MA improved impact toughness, but aEPR-g-MA decreased overall composite strength. Combining PE-g-MA and EPR-g-MA at specific concentrations produced the best balance of strength and toughness, according to the study's findings, which were supported by dynamic mechanical analysis and morphology observations³. In order to prepare and characterize bamboo-based nanocellulose for possible nanocomposite applications, Yogendra Kumar Verma et al. employed high-speed ball milling. Crystallinity alterations, a decrease in particle size to around 80 nm, and an increase in cellulose content were all verified by structural and morphological investigations (FTIR, XRD, SEM, AFM, and TEM). The relevance of the ideal filler content for better mechanical characteristics was highlighted by tensile testing of bamboo nanoparticle-epoxy biocomposites, which showed higher tensile strength up to 1.5 weight% loading, with a drop at 2 weight% due to increasing agglomeration⁴. Using profile extrusion, Junjie Xu et al. examined ultra-highly filled bamboo fiber/polypropylene composites (UH-BPCs) with filler proportions varying from 75 weight% to 90 weight%. According to experimental data, adding more filler increased modulus and creep resistance up to 85 weight% while decreasing tensile, flexural, and impact strengths. With tensile strengths of 22.39 MPa, tensile modulus of 6.05 GPa, flexural strengths of 39.86 MPa, flexural modulus of 7.19 GPa, and impact strengths of 4.22 kJ/m², UH-BPCs with moderate melt flow index (MFI) polypropylene demonstrated the best mechanical properties at this ideal filler level. These results offer important information for creating high-performance WPCs⁵. Using walnut shell powder (WSP) as a filler and partly delignified bamboo fibre (DBF) as reinforcement in a melamine-hexamethylenediamine-urea (MHU) resin matrix, Le Li et al. studied the creation of high-performance biomass-based polymer composites. According to experimental data, the composites outperformed commercial wood-plastic and natural wood-based composites in terms of flexural and tensile strengths, which reached 220 MPa and 120 MPa, respectively. The composites also demonstrated exceptional flame retardancy (UL 94-V0), high hardness (Shore D > 90), and a distinctive silvery-grey fluorescence under UV light, underscoring its potential for sustainable engineering applications with additional functional features⁶. The impact of titanium oxide (TiO₂) particles on the mechanical, tribological, water-resistance, and physical characteristics of epoxy composites reinforced with bamboo fibre and treated with 5% NaOH was examined by Ravikantha Prabhu et al. Tensile characteristics peaked at 6 weight% TiO₂, impact strength peaked at 8 weight%, and flexural strength peaked at 2 weight%, according to experimental data. The addition of TiO₂ significantly decreased water absorption and enhanced wear resistance. The work contributes to the development of new composite materials by highlighting the potential of TiO₂ as a suitable filler material to improve the overall performance of epoxy composites reinforced with bamboo fibres⁷. Mohammed Abdulkedir Alfeki et al. looked at how chopped glass fibre (CGF) filler affected the mechanical, thermal, and moisture-absorbing characteristics of polyester composites reinforced with bamboo fibres. A 40 weight% unidirectional 0° bamboo fibre composite's ultimate tensile strength (131.22 MPa), flexural strength (128.76 MPa), impact strength (113.3 kJ/m²), water absorption (1.94%), and thermal stability (255 °C, a 10% increase) were all markedly enhanced by the addition of 5 weight% CGF, according to the experimental results. The study offers statistical models for forecasting material performance and emphasises the potential of CGF-filled bamboo fibre composites for improved structural applications, especially in wall partition systems⁸. In order to improve mechanical and thermal qualities, Krushna Gouda et al. looked into creating micro bamboo filler/epoxy hybrid composites with different weight percentages of graphene nanoplatelets (GNPs). In contrast to clean or bamboo-filled epoxy composites, experimental results demonstrated that adding GNPs enhanced tensile and flexural characteristics as well as loss and storage modulus. XRD verified the presence of filler, SEM verified even distribution, and thermomechanical investigation showed enhanced thermal conductivity with a somewhat higher rate of corrosion. The results indicate that because of their improved interfacial bonding and structural performance, these hybrid composites may find use in circuit boards, electronic packaging, and thermal interface materials⁹. With and without alkali treatment (5% NaOH), Anu Gupta et al. examined the impact of ceramic fillers (AlO₃ and SiC) on the chemical and water resistance of epoxy composites supplemented with short bamboo fibres. According to experimental results, adding fillers reduced water absorption and greatly increased chemical resistance against acetic acid, NaOH, NH₄OH, Na₂CO₃, benzene, toluene, and carbon tetrachloride. Furthermore, as compared to untreated composites, alkali-treated composites demonstrated better water and chemical resistance. According to the results, bamboo fiber-reinforced composites' performance and longevity in a range of applications can be enhanced by the use of ceramic fillers and fibre treatment¹⁰. The mechanical, thermal, and morphological characteristics of hybrid nanocomposites composed of a bidirectional bamboo fibre mat, multi-walled carbon nanotubes (MWCNTs) (0.5% and 1%) in an epoxy matrix, and rice husk particles (5%, 10%, and 15%) were examined by S. Ramu et al. According to experimental data, the composite containing 0.5% MWCNTs had the maximum impact strength while the composite containing 1% MWCNTs had the highest tensile (42.66 MPa) and flexural (54.72 MPa) strengths. The bamboo fiber-rice husk composite (without MWCNTs) showed a Shore-D hardness of 84 and a thermal conductivity of 0.2577 W/mK. MWCNT-reinforced composites showed improved fiber-matrix adhesion, with less voids and fibre pull-out, according to SEM examination¹¹. The impact of maleated polypropylene (MAPP) and bamboo fibre (BF) on the mechanical, thermal, and water resistance characteristics of polypropylene (PP) composites were investigated by Sun-Young Lee et al. MAPP increased fibre dispersion, tensile strength, and water resistance, but BF decreased tensile strength and crystallinity while increasing tensile modulus, impact resistance, and creep resistance. According to the results, PP composites are optimised for better performance in structural applications by BF and MAPP changes¹².

A significant study gap on wettability investigations and their effects on void formation and tensile performance in filler-reinforced natural fibre epoxy composites was identified by the literature survey. In order to close this gap, a L_{27} orthogonal array was used to disperse Silicon Nitride (Si_3N_4) fillers in bamboo fiber-reinforced epoxy composites. The process parameters included varying the amount of Si_3N_4 filler, stirring time, and stirring speed. In order to identify the ideal parameter settings and compare experimental results with predictions, the Combined Compromise Solution (COCOSO) method in conjunction with an Artificial Neural Network (ANN) was utilised to analyse the response parameters, including void formation, contact angle, and tensile strength.

Materials and methods

Fabrication of Si_3N_4 filler reinforced bamboo fiber epoxy composites

For making the hybrid composites, Araldite LY-556 is the epoxy resin (EP) and Aradur HY-951 is the hardener. The mechanical and physical characteristics of Si_3N_4 particles and bamboo fiber vary significantly. Si_3N_4 has a density of 3.17 g/cm^3 and exists in a particulate form with a particle size of $36 \mu\text{m}$, while bamboo fiber has a lower density of 0.91 g/cm^3 and is available in a plain-woven structure with a fiber size ranging from 50 to $200 \mu\text{m}$. Bamboo fiber exhibits a tensile strength of 503 MPa , an elongation at break of 1.4% , and a tensile modulus of 35.91 MPa^{13} , whereas these properties are not applicable to Si_3N_4 particles. Figure 1(a–d) depicts a schematic of the composite manufacturing process that creates bamboo fiber/ Si_3N_4 /epoxy hybrid composites using the hand layup method. The composites were fabricated using a combination of 27 hand layup process parameters. To create a uniform dispersion, the Si_3N_4 particles are mechanically mixed with the epoxy resin at room temperature. The mixture is then thoroughly mixed with a selected stirring speed and stirring duration to ensure uniform dispersion, followed by the addition of a 10:1 stoichiometric hardener¹⁴. The Si_3N_4 epoxy solution is then applied to each layer of woven bamboo fibre using a spreader. The mould is cleaned and lined with a polyester covering to get it ready. Wax is applied to the film's edges to prevent it from adhering. A roller is used to eliminate air bubbles from the initial bamboo layer after the prepared epoxy solution has been applied. Each additional layer of bamboo fibre is inserted in the same manner with Si_3N_4 filled epoxy solution in between to form a composite with four layers.

Experimental design

Three crucial factors such as filler %, stirring duration, and stirring speed affect the performance of Si_3N_4 filler-reinforced bamboo fibre epoxy composites created by hand layup. As indicated in Table 1, each parameter was investigated at three levels using a Taguchi experimental design with an L_{27} orthogonal array in order to optimise response characteristics. Important reaction metrics were void formation, contact angle, and tensile performance. The Taguchi method's "smaller the better" criterion was used to assess void formation and contact angle, while the "larger the better" criterion was used to analyse tensile performance. The Artificial Neural Network (ANN) approach was used to perform the prediction, and the Combined Compromise Solution (COCOSO) method was used to further analyse the optimisation findings in order to provide reliable and accurate results.

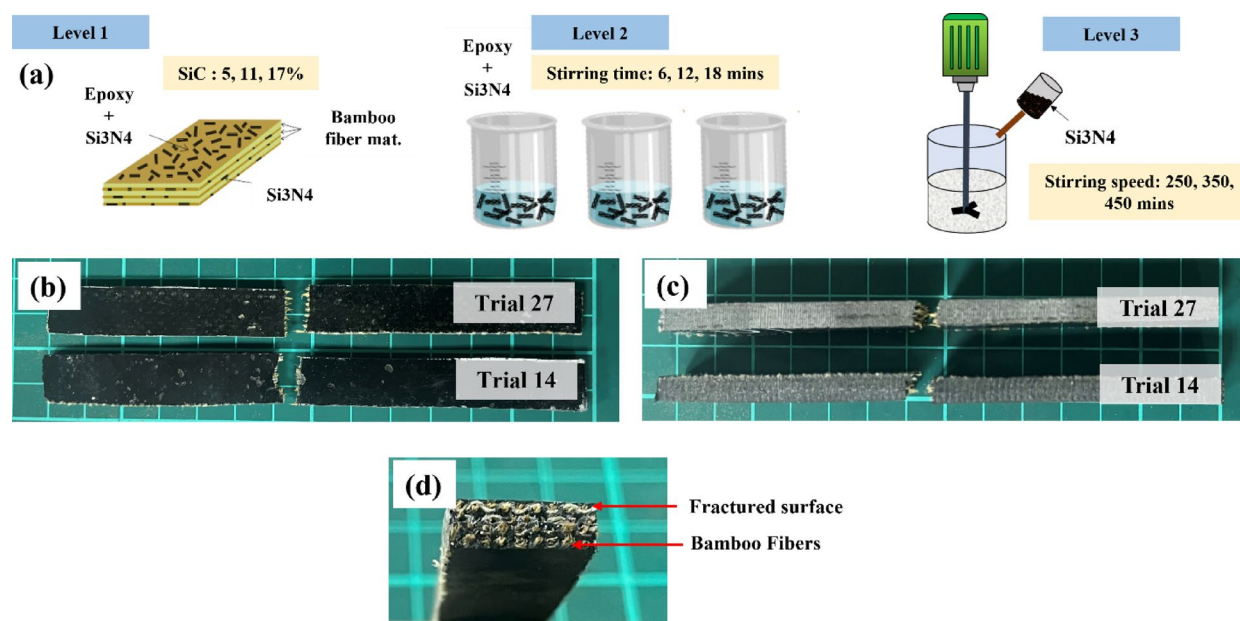


Fig. 1. (a) Schematic representation of fabrication of Si_3N_4 filler reinforced Bamboo fiber epoxy composites based on different combination of parameter settings, (b) top view and (c) side view of the fractured specimens based on optimal and un-optimal settings, and (d) sectional view of the fractured specimens.

Control factors	Level 1	Level 2	Level 3
Si ₃ N ₄ (%)	3	9	15
String Time (min)	6	12	18
Stirring Speed (rpm)	250	350	450

Table 1. Factors and levels of hand layup process parameters.

Response characteristics

The dry weight wet weight technique was used to determine the experimental densities (Ed) of the composite samples. Five samples’ average densities were used to get the average experimental density. Equation (1) was used to get the theoretical density:

$$\text{Theoretical density } (\rho_{td}) = \frac{(w_r + w_{bf} + w_s)}{\left(\frac{w_r}{\rho_r} + \frac{w_{bf}}{\rho_{bf}} + \frac{w_s}{\rho_s}\right)} \tag{1}$$

where the theoretical density is represented in g/cm³, and “w” and “ρ” stand for weight (in grammes) and density, respectively. The letters “r” also represent epoxy resin, “bf” bamboo fibre, and “S” Si₃N₄ particles. Equation (2) was used to determine the composite specimens’ void contents:

$$\text{Void content } (\%) = \frac{100 (T_d - E_d)}{T_d} \tag{2}$$

The data physics-contact angle system OCA (LMS instruments) was used to measure the water contact angle. About 5 μL of distilled water was applied to the specimens’ surface as a sessile drop in order to measure the contact angle. The wettability of the samples was investigated by measuring an average of 6–10 contact angle values. With a load cell of 2000 kgf, the Cometeq QC-506M1-204 universal testing apparatus was used to measure the composites’ tensile properties. The ASTM D3039 standard, which measures 120 × 12 × 3 mm, was used to measure the sample’s dimensions. The test was conducted using a 50 mm gauge length and a crosshead speed of 5 mm/min. Five different samples were used for the test, and the average tensile properties were calculated and reported for repeatability.

Results and discussion
SN ratio and ANOVA analysis

Based on the interaction plots for the signal-to-noise (SN) ratios, the influence of Si₃N₄ content, stirring time, and stirring speed on void percentage, contact angle, and tensile strength exhibits a clear increasing and decreasing trend. Table 2 shows the experimental reactions according to L27 OA’s experimental design. An ideal set of parameters is shown in Figs. 2(a, b), where the SN ratio adheres to a “smaller is better” principle for contact angle and void %. When the Si₃N₄ concentration is increased from 3 to 6%, the void percentage first falls, however, when it is increased to 9%, more voids are formed. Similar to this, stirring for up to 12 min improves filler dispersion and lowers void content, but, after that, void formation rises, most likely as a result of increased viscosity and air entrapment¹⁵. A similar trend is seen with stirring speed, an increase from 250 to 350 rpm leads to lower voids due to better mixing, but beyond 350 rpm, voids increase again, possibly due to turbulence¹⁶. The contact angle exhibits a comparable trend, showing a decreasing trend up to 6% Si₃N₄ and 12 min stirring time before increasing, indicating that beyond a certain threshold, poor wettability results from filler agglomeration. The stirring speed also follows this behavior, where an increase to 350 rpm enhances wettability, but beyond this point, the contact angle increases due to inadequate wetting. On the other hand, Fig. 2(c), which employs a “larger is better” criteria for tensile strength, shows that tensile strength first increases as SiN₄ concentration rises to 6% because of improved reinforcement, but then falls to 9% because particle aggregation weakens the matrix. Similarly, up to 12 min of stirring duration improves tensile strength; beyond that, too much swirling degrades the matrix. The similar pattern can be seen in stirring speed: increasing it to 350 rpm improves mechanical qualities, while exceeding 450 rpm lowers tensile strength because of fiber breaking and void formation. Overall, the results suggest that optimal mechanical performance is achieved at 6% Si₃N₄, 12 min stirring time, and 350 rpm stirring speed, beyond which detrimental effects such as void formation, reduced wettability, and weakened tensile properties dominate¹⁷. The impact of different bamboo micro filler contents on the mechanical and water-absorbing characteristics of Sisal-Bamboo-E-glass fiber-reinforced epoxy hybrid composites was examined by Gurmu et al.¹⁸. While composites with 6% bamboo micro filler showed the highest water absorption, the ideal composition—15% sisal, 10% glass, 15% bamboo fibre, 57% epoxy, and 3% bamboo micro filler—achieved the highest tensile strength (87 MPa), flexural strength (77 MPa), impact energy (8.9 J), and toughness (24.64 J/cm²). The results show that the weight percentages of fibre and filler have a major effect on composite performance, which qualifies them for structural uses in the aerospace and automotive sectors¹⁸.

Based on response Table 3 it can be found that reinforcement percentage act as the major influencing parameter followed by Stirring speed and stirring time has least effect. Based on the ANOVA table (Table 4), it can be observed that the P-values are less than zero, confirming that each parameter significantly affects the

Trial	Si ₃ N ₄ (%)	Stirring time (min)	Stirring Speed (rpm)	SNR Void formation	SNR Contact angle	SNR Tensile Strength
1	3	6	250	-11.75	-37.11	34.62
2	3	6	350	-8.13	-36.50	36.89
3	3	6	450	-12.59	-37.35	34.00
4	3	12	250	-8.50	-36.64	36.61
5	3	12	350	-4.40	-35.82	37.57
6	3	12	450	-9.71	-36.74	35.96
7	3	18	250	-12.32	-37.19	34.18
8	3	18	350	-8.91	-36.70	36.39
9	3	18	450	-13.22	-37.42	33.75
10	6	6	250	-6.24	-36.08	37.36
11	6	6	350	-2.48	-35.59	37.92
12	6	6	450	-6.73	-36.15	37.25
13	6	12	250	-1.94	-35.48	38.06
14	6	12	350	-1.06	-35.36	38.17
15	6	12	450	-3.35	-35.68	37.86
16	6	18	250	-6.93	-36.22	37.12
17	6	18	350	-3.69	-35.74	37.73
18	6	18	450	-7.38	-36.34	36.99
19	9	6	250	-13.37	-37.55	33.37
20	9	6	350	-10.26	-36.93	35.30
21	9	6	450	-14.52	-37.82	33.06
22	9	12	250	-9.97	-36.84	35.68
23	9	12	350	-4.96	-35.90	37.49
24	9	12	450	-11.22	-37.06	34.98
25	9	18	250	-14.35	-37.67	33.20
26	9	18	350	-11.00	-37.00	35.06
27	9	18	450	-14.78	-37.88	32.58

Table 2. Experimental reactions according to L27 oa's experimental design.

output response. Among the parameters, the reinforcement weight has the highest contribution, accounting for 57.55% in governing the contact angle, 56.67% for tensile strength, and 53.02% in influencing void percentage. The remaining two parameters have a more or less equal influence on the output response. Notably, stirring time makes the second-largest contribution to the output response, with 19.52% for void percentage, 19.27% for contact angle, and 18.72% for the tensile strength of the developed composite.

Artificial neural networks (ANN) modelling

Mathematical model is formulated using linear regression, as indicated from Eq. 3 to 5 in order to predict the output response. This model is developed by correlating the input and output responses attained from the experimental trails. Figure 3(a-c) shows the residual plots for voids (%), contact angle (°) and tensile strength (MPa).

$$\text{Void\%} = 1.41454 + 0.120185 \times A + 0.0219444 \times B + 0.00172778 \times C \quad (3)$$

$$\text{Contactangle} = 61.4972 + 0.490741 \times A + 0.0787037 \times B + 0.0075 \times C \quad (4)$$

$$\text{Tensilestrength} = 76.963 - 1.0963 \times A - 0.173981 \times B - 0.0135111 \times C \quad (5)$$

Artificial Neural Networks (ANN) has better efficiency in foreseeing the defects and strength of composite materials owing to its aptitude to modelling and also to derive a non-linear relationship among control factor and the selected output parameters¹⁹. By selecting the composite fabrication parameters such as stirring speed, stirring time along with the material parameter such as reinforcement percentage as input factors. ANN will help to prediction of output response such as tensile strength, contact angle and void %. In this research NFOOL is used develop ANN model, among 27 datasets 70% of trails are selected for training and remaining data variables are used for testing and validation. In network module 10 hidden layers with Levenberg- Marquardt training algorithm is selected. In the developed model the training $R = 0.9905$, validation and test $R = 0.9842$ and 0.9826 and overall $R = 0.9881$ (Fig. 4(a-d)). These performance metrics depict the better reliability and degree of accuracy of the developed model. The R values of 0.99 in training module signify the arguments between the actual and predicted data sets. Likewise the testing and validation model shows 0.9826 and 0.9842 R values that implies the better ability of developed model. Overall R value of 0.9981 further depict the capability of ANN model to predict the output variables. The achieved mean squared error (MSE) of 28.6264 at fourth epoch implies that the

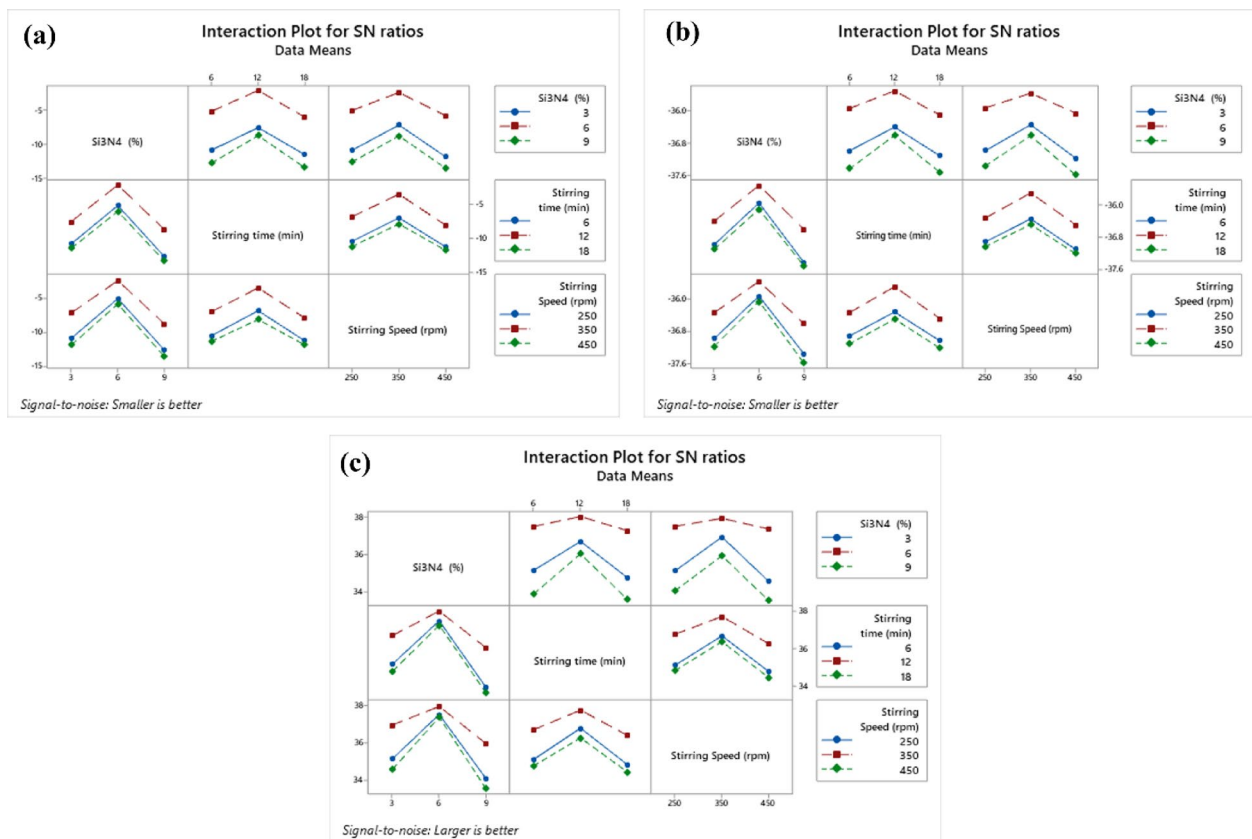


Fig. 2. Interaction plot for SN ratio for (a) void % (b) contact angle and (c) Tensile strength.

Response	Level	Si ₃ N ₄ (%)	Stirring time (min)	Stirring Speed (rpm)
Voids (%)	1	−9.948	−9.563	−9.484
	2	−4.421	−6.123	−6.101
	3	−11.604	−10.287	−10.388
	Delta	7.182	4.164	4.288
	Rank	1	3	2
Contact Angle (°)	1	−36.83	−36.79	−36.75
	2	−35.85	−36.17	−36.17
	3	−37.18	−36.91	−36.94
	Delta	1.34	0.74	0.77
	Rank	1	3	2
Tensile Strength (MPa)	1	35.55	35.53	35.58
	2	37.61	36.93	36.95
	3	34.53	35.22	35.16
	Delta	3.08	1.71	1.79
	Rank	1	3	2

Table 3. Response table of dependent variables.

model has converged to lower error values and implies it better predictability (Fig. 5). Figure 6(a-c) shows the comparison of experimental and predicted values which witnessed that the obtained ANN predicted values were more coherent with the observed experimental results.

Correlation of combined parametric impacts on dependent variables

Based on the 3D surface plots (Fig. 7(a-i)), the impact of Si₃N₄ filler percentage, stirring time, and stirring speed on void content, contact angle, and tensile strength of the epoxy composite reinforced with woven bamboo fibers can be analyzed quantitatively. From Figs. 7(a-c), void percentage generally decreases with moderate levels of

Parameter	Source	DF	Adj SS	Adj MS	F	P	Contribution %
Voids (%)	Si ₃ N ₄ (%)	2	24.548	12.2740	85.51	0.000	53.02
	String time (min)	2	9.056	4.5282	31.55	0.000	19.56
	String Speed (rpm)	2	9.821	4.9104	34.21	0.000	21.21
	Error	20	2.871	0.1435			
	Total	26	46.296				
Contact Angle (°)	Si ₃ N ₄ (%)	2	519.63	259.814	152.05	0.000	57.55
	String time (min)	2	171.50	85.748	50.18	0.000	18.99
	String Speed (rpm)	2	177.61	88.803	51.97	0.000	19.67
	Error	20	34.18	1.709			
	Total	26	902.91				
Tensile Strength (MPa)	Si ₃ N ₄ (%)	2	2282.1	1141.07	85.75	0.000	56.67
	String time (min)	2	724.7	362.36	27.23	0.000	18.00
	String Speed (rpm)	2	753.9	376.94	28.33	0.000	18.72
	Error	20	266.1	13.31			
	Total	26	4026.9				

Table 4. ANOVA of dependent variables.

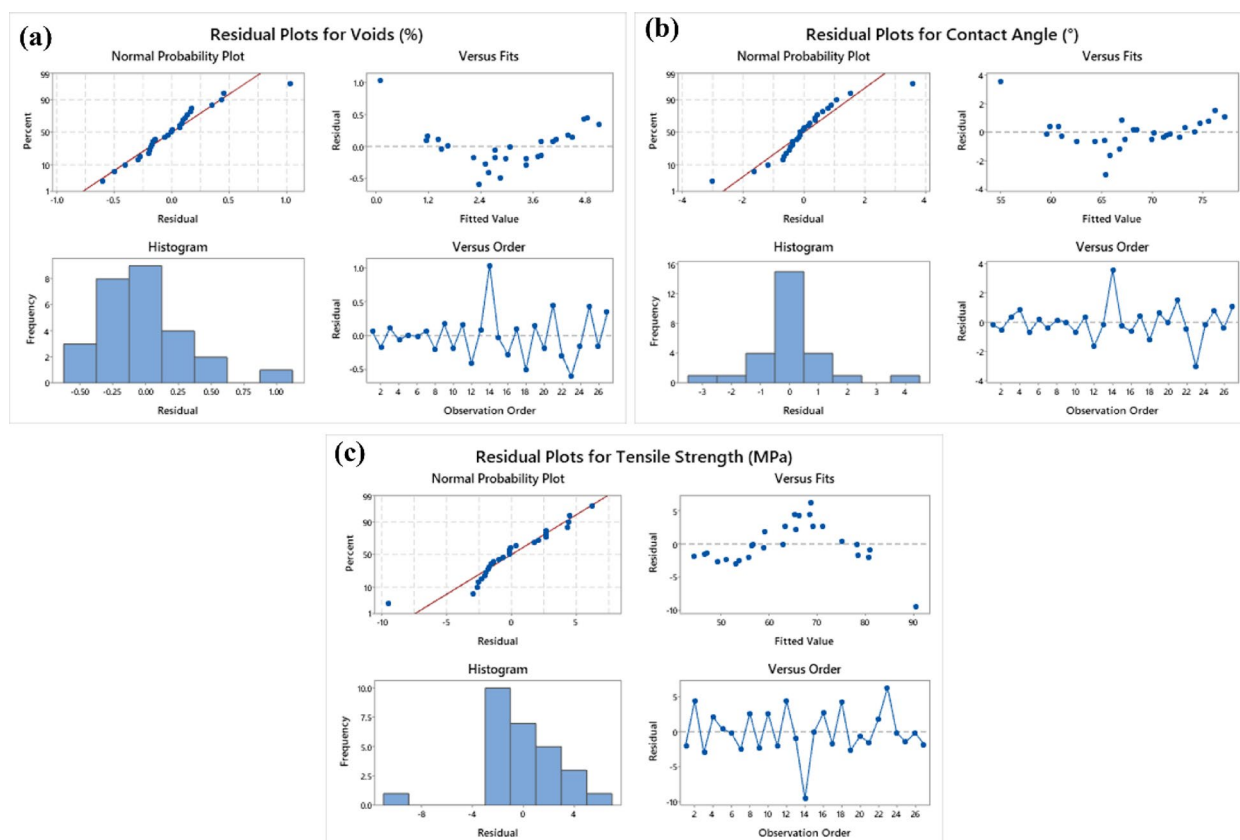


Fig. 3. Residual plots for (a) voids (%), (b) contact angle (°) and (c) tensile strength (MPa).

Si₃N₄ (6%) and controlled stirring conditions. At 3% Si₃N₄, voids are higher (~ 5.5–6%), whereas at 6% Si₃N₄ with 12 min stirring time and 350 rpm, voids drop to ~ 3.5%, indicating improved matrix-filler interaction and reduced agglomeration. However, beyond 6% Si₃N₄ (at 9%), void content increases again (~ 5%), likely due to particle clustering and reduced resin flowability. Stirring time of 6 min at 250 rpm results in higher voids (~ 5.8%), whereas at 12 min and 350 rpm, voids reduce significantly (~ 3.5%). Excessive stirring (18 min at 450 rpm) leads to increased void content (~ 4.5%), suggesting that high mechanical agitation induces air entrapment and turbulence. Figure 7(d–f) depict the effect of process parameters on wettability, represented by the contact angle. The contact angle reduces as Si₃N₄ increases from 3 to 6% (from ~ 72° to ~ 66°), confirming enhanced filler dispersion and surface energy. However, at 9% Si₃N₄, contact angle increases (~ 74°), indicating

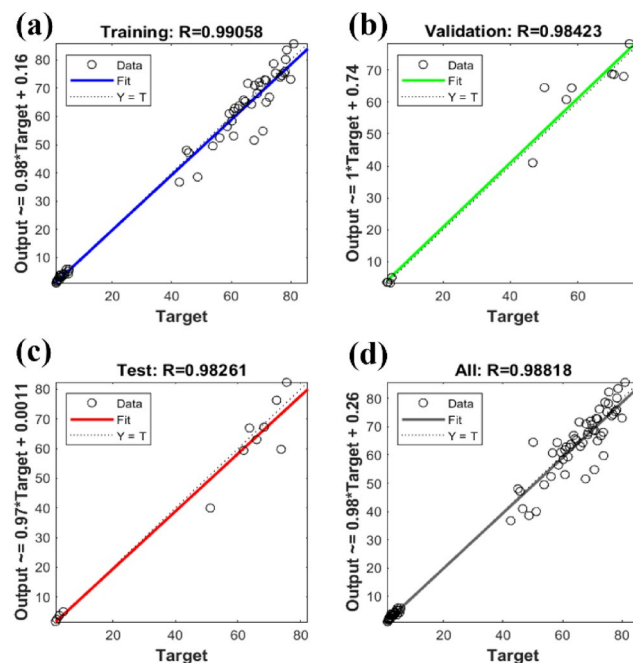


Fig. 4. Regression plot of datasets (a) Training (b) Validation (c) Test and (d) All.

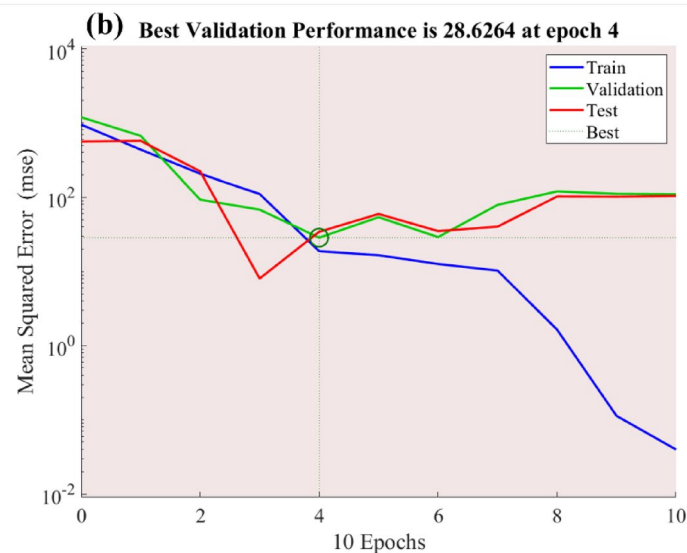
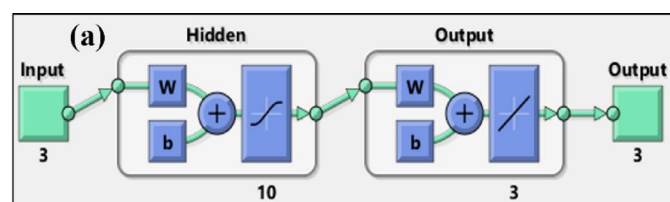


Fig. 5. (a) Artificial Neural Network modelling and its corresponding (b) performance validation results.

poor wettability due to particle agglomeration. Stirring time of 6 min shows the highest contact angle ($\sim 75^\circ$), suggesting incomplete dispersion, while 12 min at 350 rpm leads to optimal wettability ($\sim 66^\circ$). Beyond 12 min (at 18 min, 450 rpm), the contact angle increases ($\sim 72^\circ$), implying excessive stirring negatively impacts interface homogeneity. Tensile strength trends (Fig. 7(g-i)) reveal an optimum Si_3N_4 content of 6%, yielding the highest tensile strength (~ 80 MPa) at 12 min stirring time and 350 rpm. At 3% Si_3N_4 , tensile strength is lower (~ 70

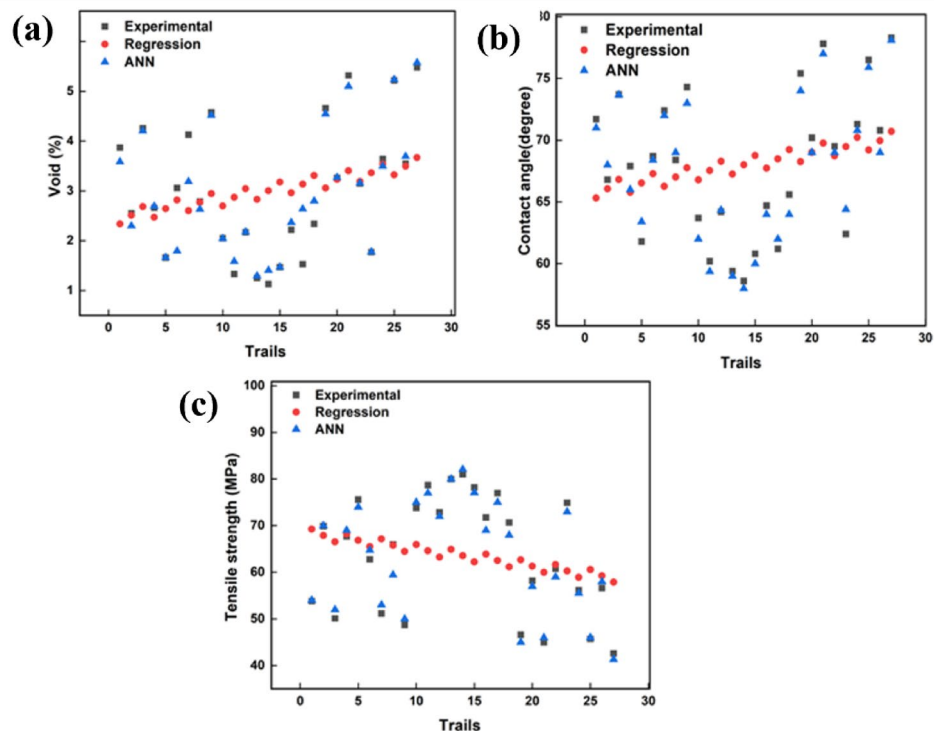


Fig. 6. Experimental vs. predicted values for (a) Void % (b) Contact angle and (c) Tensile strength based on ANN modeling.

MPa), likely due to insufficient reinforcement. Beyond 6%, at 9% Si_3N_4 , tensile strength drops (~ 68 MPa) due to excessive filler causing stress concentrations and brittleness. During the fracture propagation analysis, it was witnessed that the brittle like fracture was observed at the epoxy region and ductile like elongated fibers were observed in the bamboo fiber region. Figure 8 (a-c) depicts the fracture propagation along the epoxy region and the bamboo fiber region. Stirring speed plays a crucial role; at 250 rpm, tensile strength remains low (~ 68 MPa) due to inadequate mixing, whereas at 350 rpm, tensile strength peaks (~ 80 MPa), showing well-dispersed reinforcement. However, at 450 rpm, tensile strength drops (~ 72 MPa), indicating over-shearing of fibers and resin weakening. Similarly, stirring time follows an optimum trend where 6 min results in lower tensile strength (~ 70 MPa), 12 min reaches a peak (~ 80 MPa), and 18 min reduces tensile strength (~ 72 MPa). These findings confirm that the best mechanical performance is achieved at 6% Si_3N_4 , 12 min stirring time, and 350 rpm stirring speed, balancing void reduction, wettability, and tensile strength.

Multi-Criteria Decision-Making using COCOSO approach

Multi-Criteria Decision-Making (MCDM) approach are adopted by researchers and industrial expert to attain single solution for multi response problems²⁰. Combined Compromise Solution (COCOSO) method is one among the MCDM method. It integrates several combination strategies for attaining accurate and balanced solution. Compared to traditional approach such as TOPSIS which works on ideal solutions concepts, COCOSO works with subjective criteria weighting that lead to better ranking²¹. Likewise Grey Relational Analysis focus in on relational coefficients to attained optimal solution while COCOSO provides comprehensive evaluation for making decisions²². Other approaches such as AHP will struggle with pairwise comparisons with input variables that can be simplified without compromising the reliability. These facts will help in handling complex decision at minimal time duration while compared with other conventional methods. The step involved in COCOSO approach is as follows.

Step 1: Normalization of output response: The attained data set are formulated to a decision matrix and their values are standardize between 0 and 1 by means of normalization. Two equations are used for normalization based on it criteria, minimal range of void % and contact angle (Eq. 6) gives better properties to composite hence smaller the better is selected; similar higher tensile strength are needed for composite hence larger the better is selected for tensile strength (Eq. 7). The attained values are shown in Table 5.

$$Nm_{ij} = \frac{max_{aij} - a_{ij}}{max_{aij} - min_{aij}} \quad (6)$$

$$NM_{ij} = \frac{a_{ij} - min_{aij}}{max_{aj} - min_{aij}} \quad (7)$$

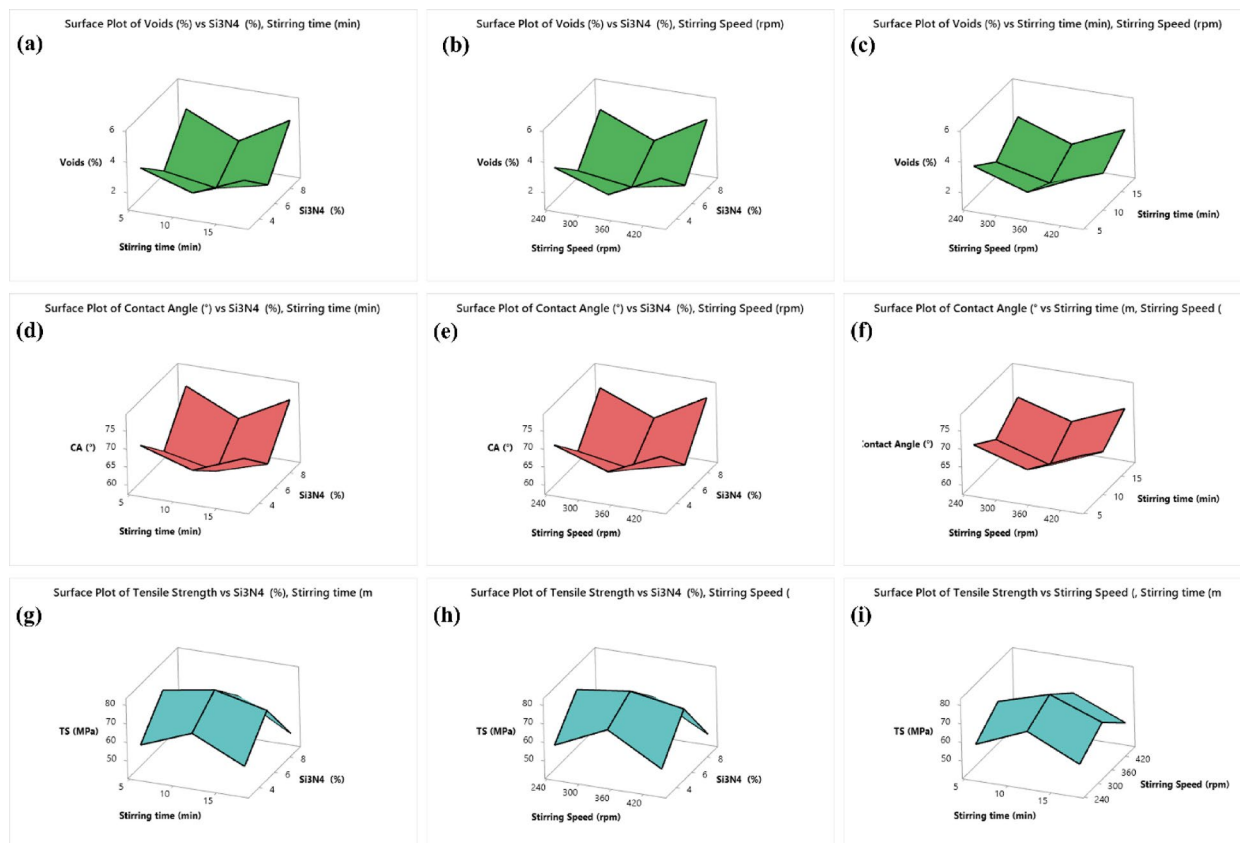


Fig. 7. 3D surface plot based on the combined effect of process parameter on output response (a-c) Voids, (d-f) Contact angle and (g-i) Tensile strength.

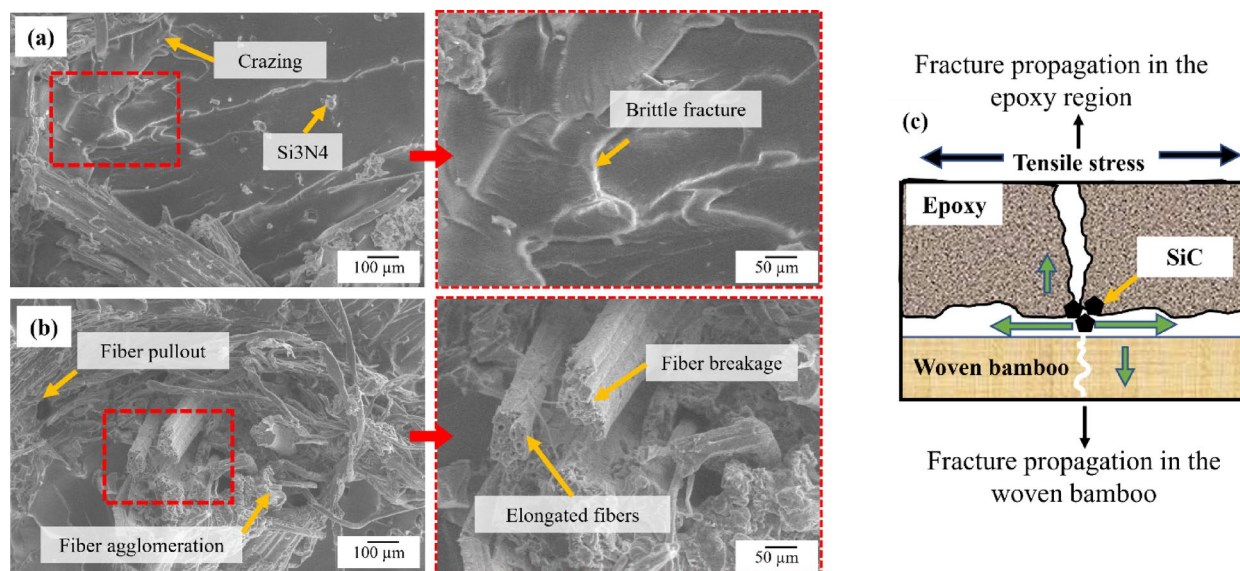


Fig. 8. Fracture propagation analysis on the (a) epoxy region and (b) bamboo fiber region and (c) its corresponding schematic representation of fracture propagation.

Trail	Normalized Void %	Normalized CA	Normalized TS	Weighted sum of void %	Weighted sum of CA	Weighted sum of TS
1	0.3701	0.3350	0.2921	0.1234	0.1117	0.0974
2	0.6736	0.5838	0.7105	0.2245	0.1946	0.2368
3	0.2805	0.2335	0.1959	0.0935	0.0778	0.0653
4	0.6483	0.5279	0.6540	0.2161	0.1760	0.2180
5	0.8782	0.8376	0.8587	0.2927	0.2792	0.2862
6	0.5563	0.4873	0.5255	0.1854	0.1624	0.1752
7	0.3103	0.2995	0.2235	0.1034	0.0998	0.0745
8	0.6184	0.5025	0.6087	0.2061	0.1675	0.2029
9	0.2069	0.2030	0.1595	0.0690	0.0677	0.0532
10	0.7885	0.7411	0.8124	0.2628	0.2470	0.2708
11	0.9540	0.9188	0.9394	0.3180	0.3063	0.3131
12	0.7609	0.7157	0.7882	0.2536	0.2386	0.2627
13	0.9724	0.9594	0.9732	0.3241	0.3198	0.3244
14	1.0000	1.0000	1.0000	0.3333	0.3333	0.3333
15	0.9218	0.8883	0.9266	0.3073	0.2961	0.3089
16	0.7494	0.6904	0.7594	0.2498	0.2301	0.2531
17	0.9080	0.8680	0.8949	0.3027	0.2893	0.2983
18	0.7218	0.6447	0.7313	0.2406	0.2149	0.2438
19	0.1885	0.1472	0.1048	0.0628	0.0491	0.0349
20	0.5103	0.4112	0.4063	0.1701	0.1371	0.1354
21	0.0368	0.0254	0.0627	0.0123	0.0085	0.0209
22	0.5356	0.4467	0.4742	0.1785	0.1489	0.1581
23	0.8529	0.8071	0.8413	0.2843	0.2690	0.2804
24	0.4230	0.3553	0.3525	0.1410	0.1184	0.1175
25	0.0598	0.0914	0.0812	0.0199	0.0305	0.0271
26	0.4437	0.3807	0.3652	0.1479	0.1269	0.1217
27	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 5. Normalized values of comparability matrix.

Max a_{ij} - maximum values of a_{ij} ; Min a_{ij} is the minimal value; NM $_{ij}$ is the normalization of smaller the better and NM $_{ij}$ is the normalization of larger the better.

Step 2: Calculation of weightage for normalized decision matrix: In this research equal weightage is given for all the output response. The weightage of the sum (Wsum) and product (Wprd) can be computed using Eqs. (8) and (9) where $E = 1/3$, the attained values are depicted in Tables 5 and 6.

$$W_{\text{sum}} = \sum_{j=1}^n NM_{ij} \times E \quad (8)$$

$$W_{\text{prd}} = \sum_{j=1}^n NM_{ij}^E \quad (9)$$

$$W_{\text{prd}} = \sum_{j=1}^n NM_{ij}^E \quad (9)$$

Step 3 computing the appraisal score There are three individual valuation grading systems adopted for various possibilities of relative weights as depicted in Eq. 10 to 13. In Eq. (13), the value of λ is considered as 0.5. However, the minimum sum and product of weighted matrix is zero (Table 6) and the attained parameter scores is tabulated in Table 7. Based on COCOSO approach the optimal strength in composite can be attained by add 6 wt% of Si_3N_4 with 12 min of stirring time at 350 rpm of string speed.

$$Z_a = \frac{W_{\text{sum}} + W_{\text{prd}}}{\sum_{i=1}^m (W_{\text{sum}} + W_{\text{prd}})} \quad (10)$$

$$Z_c = \frac{\lambda (W_{\text{sum}}) + (1 - \lambda) (W_{\text{prd}})}{\lambda \max W_{\text{sum}} + (1 - \lambda) \max W_{\text{prd}}}; 0 \leq \lambda \leq 1 \quad (11)$$

$$Z_i = (Z_a \times Z_c)^{1/2} + \frac{1}{2} (Z_a \times Z_c) \quad (12)$$

Trail	Weighted product of void %	Weighted product of CA	Weighted product of TS	Sum of weighted matrix (Wsum)	Sum of product weighted matrix (Wprd)
1	0.7180	0.6945	0.6635	0.3324	2.0760
2	0.8766	0.8358	0.8923	0.6559	2.6046
3	0.6546	0.6158	0.5808	0.2366	1.8511
4	0.8655	0.8082	0.8680	0.6101	2.5417
5	0.9576	0.9426	0.9505	0.8582	2.8507
6	0.8224	0.7869	0.8070	0.5230	2.4163
7	0.6770	0.6691	0.6068	0.2778	1.9529
8	0.8520	0.7950	0.8475	0.5766	2.4945
9	0.5914	0.5878	0.5423	0.1898	1.7215
10	0.9239	0.9050	0.9331	0.7807	2.7619
11	0.9844	0.9722	0.9794	0.9374	2.9360
12	0.9129	0.8945	0.9237	0.7550	2.7312
13	0.9907	0.9863	0.9910	0.9683	2.9680
14	1.0000	1.0000	1.0000	1.0000	3.0000
15	0.9732	0.9613	0.9749	0.9123	2.9095
16	0.9083	0.8838	0.9123	0.7330	2.7045
17	0.9684	0.9539	0.9637	0.8903	2.8859
18	0.8970	0.8639	0.9009	0.6993	2.6618
19	0.5734	0.5280	0.4715	0.1469	1.5729
20	0.7991	0.7436	0.7407	0.4426	2.2834
21	0.3326	0.2939	0.3973	0.0416	1.0237
22	0.8121	0.7644	0.7798	0.4855	2.3564
23	0.9483	0.9311	0.9440	0.8338	2.8234
24	0.7507	0.7083	0.7064	0.3769	2.1654
25	0.3910	0.4504	0.4330	0.0774	1.2744
26	0.7627	0.7248	0.7148	0.3965	2.2023
27	0.0000	0.0000	0.0000	0.0000	0.0000

Table 6. Sum and product of comparability matrix.

Sensitivity analysis on effect of control factor on output response

From Table 8, it can be observed that increase in filler for level 1 to 2 decrease the void percentage of 0.55% with respect to 1% addition. It decreases and increase the tensile strength by 2.43° and 4.86 MPa. Conversely above 6% addition increase voids 0.74% for every 1% and decrease the tensile strength to 7.10 MPa attributed to the clustering effect. In case of stirring speed at the range of 250 to 350 rpm decrease voids by 1.10% for every 100 rpm and improve its tensile strength by 3.50 MPa per 100 rpm, while increase in speed decreases the composite strength. Likewise, 6 to 12 min of stirring time increase the tensile strength by 1.56 MPa per minute, conversely above 12 min increase the void by 0.41% per minute. Based on this fact it can be observed that 6% Si₃N₄, 350 rpm, and 12 min is the optimal parameter for developing composite with better strength and minimal defect.

Confirmation experiments

Void formation

The observed variation in void formation across the trials (Fig. 9(a-c)) can be attributed to the interplay between the filler material (Si₃N₄), stirring time, and stirring speed, which influence the homogeneity, viscosity, and air entrapment during the composite fabrication process. In Trial 14, the combination of 6% Si₃N₄, 12 min of stirring, and 350 rpm resulted in minimal void formation (1.13%), suggesting an optimal balance between these parameters. Similarly, the thermal and mechanical characteristics of epoxy composites supplemented with modified bamboo micron fibres, such as fibres coated with carbon nanoparticles and milled bamboo fillers, were examined by Hrishikesh Dutta et al. The findings demonstrated that whereas milled bamboo fillers improved the mechanical and glass transition behaviour, surface-coated fibres decreased water absorption. The efficiency of bamboo-based fillers in enhancing composite performance was demonstrated by the maximum tensile strength (90.258 MPa) that was attained with 5 weight% milled bamboo fillers, which represented a 93.01% improvement over plain epoxy (46.762 MPa)²³. At 6% Si₃N₄, the filler content is sufficient to enhance the matrix's properties without significantly increasing viscosity, which can hinder proper fiber wetting and lead to voids²⁴. The similar observation was presented by Yingjie Qiao et al.²⁵. The moderate stirring time of 12 min allows for adequate dispersion of Si₃N₄ and removal of entrapped air, while the stirring speed of 350 rpm provides enough shear force to ensure uniform mixing without introducing excessive air bubbles which was witnessed in the Figure. 10(b). In contrast, Trial 27, with 9% Si₃N₄, 18 min of stirring, and 450 rpm, exhibited the highest void formation (5.48%). The higher filler content likely increased the viscosity of the epoxy, making it more difficult to wet the bamboo fibers uniformly and trap air during mixing²⁶. During 18 min of stirring, prolonged agitation

Trail	Za	Zc	Z	Rank
1	0.0315	0.6021	0.3263	20
2	0.0426	0.8151	0.4462	12
3	0.0273	0.5219	0.2817	22
4	0.0412	0.7879	0.4308	13
5	0.0485	0.9272	0.5103	6
6	0.0384	0.7348	0.4007	15
7	0.0292	0.5577	0.3015	21
8	0.0401	0.7678	0.4194	14
9	0.0250	0.4778	0.2574	23
10	0.0463	0.8856	0.4865	8
11	0.0506	0.9683	0.5340	3
12	0.0456	0.8715	0.4784	9
13	0.0514	0.9841	0.5431	2
14	0.0523	1.0000	0.5523	1
15	0.0500	0.9554	0.5266	4
16	0.0449	0.8594	0.4715	10
17	0.0494	0.9441	0.5200	5
18	0.0439	0.8403	0.4606	11
19	0.0225	0.4299	0.2310	24
20	0.0356	0.6815	0.3707	17
21	0.0139	0.2663	0.1420	26
22	0.0371	0.7105	0.3870	16
23	0.0478	0.9143	0.5029	7
24	0.0332	0.6356	0.3450	19
25	0.0177	0.3379	0.1808	25
26	0.0340	0.6497	0.3529	18
27	0.0000	0.0000	0.0000	27

Table 7. Ranking and attributed appraisal scores.

Parameter	Range	Voids (%/unit)	Contact angle (°/unit)	Tensile strength (MPa/unit)
Si₃N₄	3% → 6%	−0.55	−2.43°/1%	+ 4.86 MPa/1%
	6% → 9%	0.74	+ 3.40°/1%	−7.10 MPa/1%
Stirring Speed	250 → 350 rpm	−1.10%/100 rpm	−2.50°/100 rpm	+ 3.50 MPa/100 rpm
	350 → 450 rpm	+ 0.83%/100 rpm	+ 2.47°/100 rpm	−3.89 MPa/100 rpm
Stirring Time	6 → 12 min	−0.34%/min	−1.03°/min	+ 1.56 MPa/min
	12 → 18 min	+ 0.41%/min	+ 1.25°/min	−1.87 MPa/min

Table 8. Sensitivity analysis on effect of control factor on output response.

introduced excessive air bubbles into the epoxy mixture, which became trapped and led to increased porosity. This higher porosity weakened the composite by creating stress concentration points which was also stated by Yutai Su et al.²⁷ This phenomenon was evident in the FESEM image (Fig. 9(c)) taken at the fracture region of Trial 27 coupon. The combination of these factors likely led to poor fiber-matrix adhesion and increased void content. The results highlight the importance of optimizing filler content, stirring time, and speed to achieve a homogeneous mixture with minimal voids which was also emphasized by Shuiwen Zhu et al.²⁸. Excessive filler or aggressive mixing can disrupt the composite's integrity, while insufficient mixing may lead to poor dispersion and air entrapment¹⁴. Trial 14 represents a balanced approach, where the parameters synergistically minimize void formation, ensuring better mechanical properties and performance of the composite.

Wettability analysis

The wettability analysis of the epoxy composite, as reflected by the contact angle measurements (Fig. 10(a)), plays a pivotal role in determining the interfacial adhesion between the epoxy matrix, bamboo fibers, and Si₃N₄ filler, which directly impacts the composite's mechanical performance and structural integrity. In Trial 14, the minimum contact angle of 58.6° indicates high wettability, suggesting that the epoxy matrix effectively wets the bamboo fibers and Si₃N₄ filler (Fig. 10(b)). This can be attributed to the optimal combination of 6% Si₃N₄, 12 min of stirring, and 350 rpm, which likely resulted in a well-dispersed filler and a homogeneous matrix with moderate viscosity. The moderate filler content ensures that the epoxy retains sufficient fluidity to penetrate

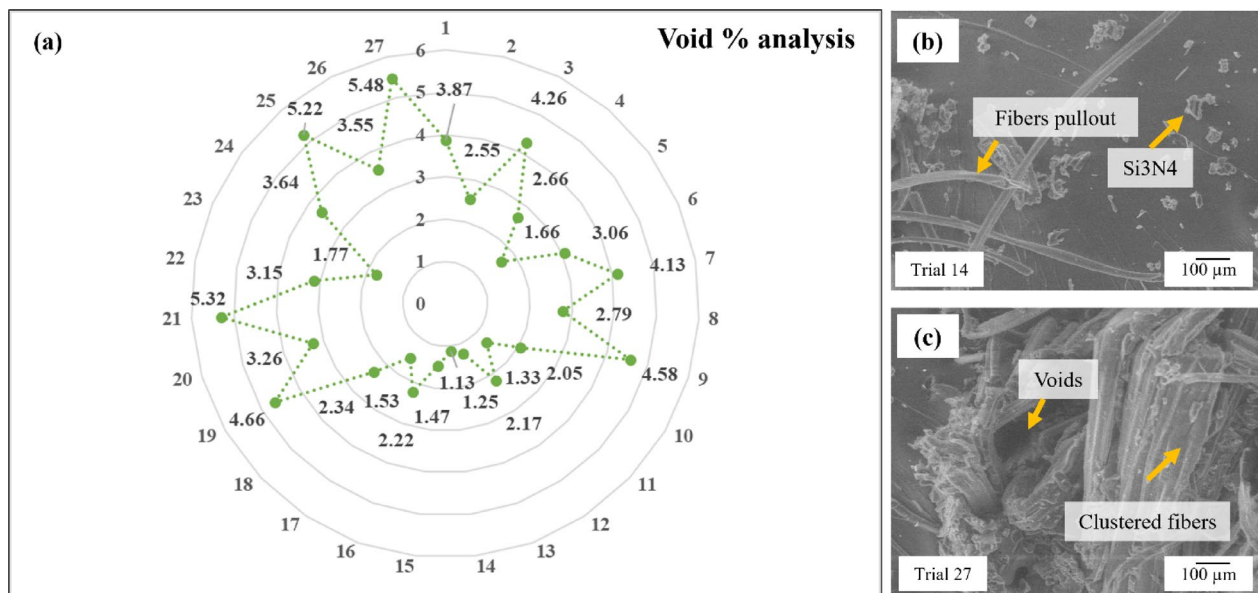


Fig. 9. (a) Void percentage according to 27 experimental trails and the FESEM image comparison of optimal (b) Trial 14 and (c) un-optimal settings (Trial 27).

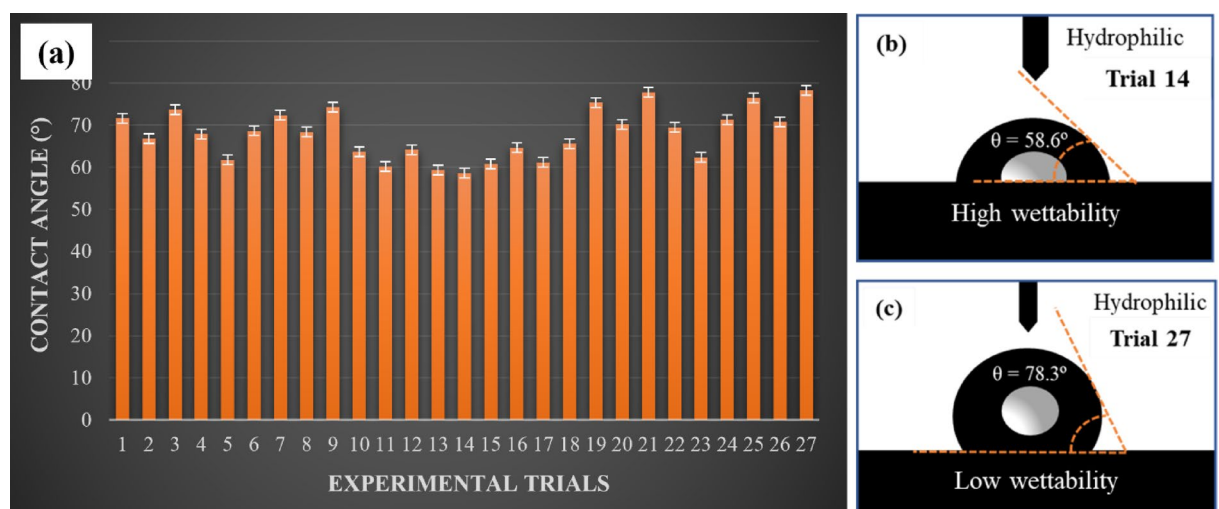


Fig. 10. Contact angle measurements according to (a) 27 experimental trails and the wettability comparison of optimal (b) Trial 14 and (c) un-optimal settings (Trial 27).

the woven bamboo fibers, while the stirring time and speed provide adequate shear forces to disperse the filler and remove entrapped air without introducing excessive turbulence²⁹. The high wettability in Trial 14 promotes strong interfacial bonding between the matrix and reinforcement, which is critical for efficient load transfer and enhanced mechanical properties such as tensile strength, flexural strength, and impact resistance³⁰. Additionally, the low void formation (1.13%) observed in Trial 14 further supports the effectiveness of this parameter combination, as good wetting minimizes air entrapment and ensures a more uniform composite structure. In contrast, Trial 27, with a maximum contact angle of 78.3°, exhibits low wettability, indicating poor wetting of the bamboo fibers by the epoxy matrix (Fig. 10(c)). This can be explained by the higher Si₃N₄ content (9%), longer stirring time (18 min), and higher stirring speed (450 rpm), which collectively increase the viscosity of the epoxy and reduce its ability to spread over and penetrate the fibers³¹. Rahul Kumar et al. used linear elastic fracture mechanics to examine the fracture characteristics of epoxy composites reinforced with micro-size bamboo particle fillers. FTIR, XRD, and TGA tests verified that alkaline treatment decreased hydrophilicity and enhanced filler compatibility. The results demonstrated the efficiency of bamboo fillers in increasing composite strength, with fracture toughness increasing by 32% to a high of 0.678 MPa·m^{0.5}³². The high filler content leads to a thicker, less fluid matrix, while the extended stirring time and aggressive speed may introduce air bubbles and cause filler agglomeration, further hindering wetting. The low wettability in Trial 27 results in

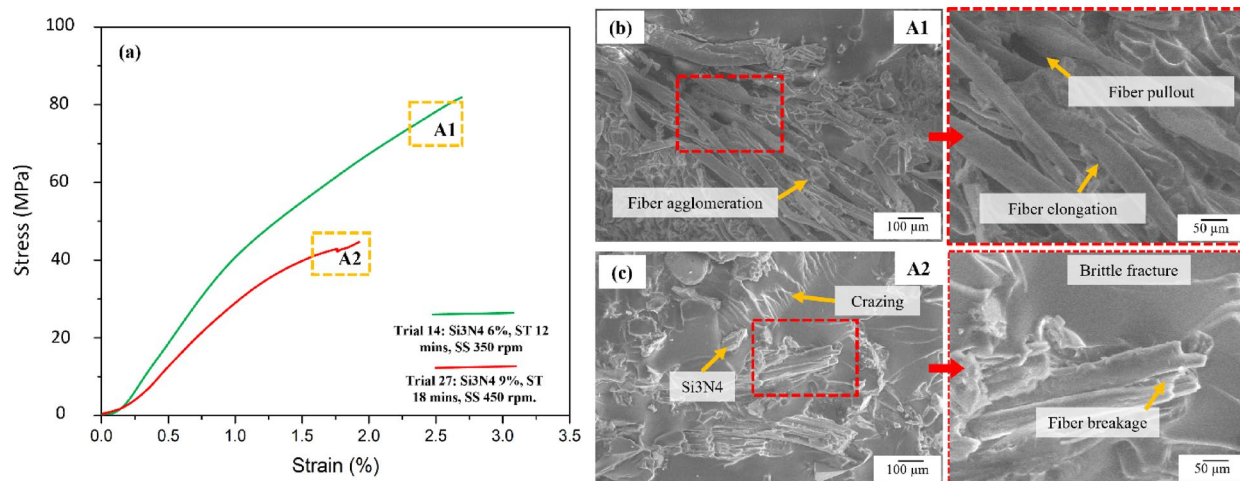


Fig. 11. (a) Stress – Strain analysis and its corresponding fracture FESEM images of (b) optimal (Trial 14) and (c) un-optimal settings (Trial 27).

weak interfacial adhesion, which can lead to stress concentrations at the fiber-matrix interface and reduce the composite's overall mechanical performance³³. Moreover, the high void formation (5.48%) observed in Trial 27 exacerbates these issues, as voids act as stress concentrators and initiation points for failure. The impact of wettability is significant, as it directly affects the composite's efficiency in transferring stress between the matrix and the reinforcement, as highlighted by Yue Wang et al.³⁴. High wettability, as seen in Trial 14, ensures strong interfacial bonding and a more homogeneous structure, leading to superior mechanical properties and durability. Conversely, low wettability, as in Trial 27, compromises interfacial adhesion and increases void content, resulting in reduced strength and potential premature failure. Therefore, optimizing wettability through careful selection of filler content, stirring time, and speed is essential for achieving a high-performance composite with balanced mechanical properties and minimal defects.

Tensile performance

The tensile performance of the epoxy composite (Fig. 11), as demonstrated by the significant difference in tensile stress between Trial 14 (81.02 MPa) and Trial 27 (42.58 MPa), can be attributed to the interplay between the process parameters (Si₃N₄ content, stirring time, and stirring speed) and their influence on wettability, void formation, and interfacial adhesion. In Trial 14, the combination of 6% Si₃N₄, 12 min of stirring, and 350 rpm likely resulted in optimal wettability, as evidenced by the minimal contact angle (58.6°) and low void formation (1.13%). The moderate Si₃N₄ content ensured that the epoxy matrix retained sufficient fluidity to effectively wet the woven bamboo fibers, promoting strong interfacial bonding³⁵. During trial 14, it was observed that, Si₃N₄ particles acted as barriers to crack growth by blocking and deflecting fracture paths, which helped in delaying crack propagation. Their strong interfacial bonding with the matrix improved energy absorption during fracture, enhancing the composite's strength which was also observed by Yijie Wang et al.³⁶. The stirring time and speed were balanced to achieve homogeneous dispersion of the filler without introducing excessive air bubbles, which further enhanced the composite's structural integrity. The strong interfacial adhesion, facilitated by good wettability, allowed for efficient load transfer between the matrix and reinforcement, leading to the observed high tensile strength (Fig. 12(b)). In contrast, Trial 27, with 9% Si₃N₄, 18 min of stirring, and 450 rpm, exhibited poor wettability, as indicated by the higher contact angle (78.3°) and increased void formation (5.48%). The higher filler content increased the viscosity of the epoxy, reducing its ability to penetrate and wet the bamboo fibers effectively. Additionally, the extended stirring time and higher speed likely introduced air bubbles and caused filler agglomeration, further degrading the fiber-matrix interface (Fig. 12(c)). The mechanical and wear characteristics of bamboo-glass-epoxy hybrid composites supplemented with TiO₂ and ZrO₂ ceramic fillers at weight fractions of 3%, 6%, and 9% are examined by P. Sneha Latha et al. The use of these fillers improves composite performance, according to experimental data from tensile and erosion wear tests; however, the effects vary according on the kind and concentration of filler. The study comes to the conclusion that inorganic fillers greatly increase the hybrid composite's mechanical strength and resistance to wear, which makes it more appropriate for structural uses³⁷. The poor wettability in Trial 27 resulted in weak interfacial adhesion, which compromised the composite's ability to transfer stress efficiently. This, combined with the high void content, created stress concentration points that significantly reduced the tensile strength³⁸. The consequences of wettability are profound, as it directly influences the mechanical performance of the composite. High wettability, as seen in Trial 14, ensures strong interfacial bonding and a uniform distribution of stress, leading to superior tensile properties. Conversely, low wettability, as in Trial 27, results in weak interfacial adhesion and increased void content, which act as failure initiation points under tensile loading³⁹. It was witnessed that, at 6% Si₃N₄, the filler content was low enough to maintain good epoxy fluidity, which allowed better wettability of the bamboo fibers, stronger interfacial adhesion, uniform filler dispersion, and minimal void formation, all of which enabled efficient stress transfer and improved tensile strength. In contrast, at 9% Si₃N₄, the excessive filler had increased

the epoxy's viscosity, reduced its wettability, caused poor fiber impregnation, promoted filler agglomeration, and increased void content, leading to weak interfacial bonding and lower tensile strength. The similar phenomenon was illustrated by Min Sun et al.⁴⁰. These findings underscore the importance of optimizing process parameters to achieve a balance between filler content, stirring time, and speed, ensuring good wettability and minimal void formation. Such optimization is critical for developing high-performance composites with enhanced mechanical properties and structural reliability.

Conclusion

This study investigated the void formation, wettability, and tensile performance of Si₃N₄-Reinforced Bamboo Fibre Epoxy Composites after they were fabricating utilizing various combinations of critical parameters such as filler %, stirring speed, and stirring duration. Through fracture surface analysis, the influence of these factors was assessed. The optimum conditions discovered utilising the COCOSO technique with ANN prediction were confirmed by verification experiments, leading to the following results.

- As Si₃N₄ content increases from 3 to 6%, the contact angle decreases from ~72° to ~66°, indicating improved filler dispersion and higher surface energy. However, at 9% Si₃N₄, the contact angle rises to ~74°, suggesting reduced wettability caused by particle agglomeration.
- While swirling for 12 min at 350 rpm considerably lowers voids (~ 3.5%), stirring for 6 min at 250 rpm resulted in a greater void content (~ 5.8%). On the other hand, excessive stirring at 450 rpm and 18 min increases void formation (~ 4.5%), suggesting that high mechanical agitation encourages turbulence and air entrapment.
- Stirring speed significantly impacts tensile strength, with 250 rpm resulting in lower strength (~ 68 MPa) due to insufficient mixing, while 350 rpm achieves the highest tensile strength (~ 80 MPa), indicating optimal reinforcement dispersion. However, at 450 rpm, tensile strength decreases (~ 72 MPa), suggesting excessive shear forces that degrade fiber integrity and weaken the resin.
- Brittle-like fractures were seen in the epoxy region during the fracture propagation study during the tensile performance analysis, whereas ductile-like elongated fibres were seen in the bamboo fibre region.
- The ideal combination of 6% Si₃N₄, 12 min of stirring, and 350 rpm stirring speed increased wettability by 25.1% and decreased void formation by 79.3%, producing a defect-free composite with improved structural integrity and superior mechanical performance, according to COCOSO experimental and analytical data combined with ANN predictions.
- The proposed research work on Si₃N₄-reinforced bamboo fiber epoxy composites is specifically applicable to lightweight structural components where high strength, durability, and eco-friendliness are critical, such as in automotive interior panels, sports equipment, construction materials, and sustainable packaging solutions. It can also be used in aerospace secondary structures and marine applications where reduced weight and enhanced mechanical performance are important.

Data availability

The experimental datasets obtained from this research work and then the analysed results during the current study are available from the corresponding author on reasonable request.

Received: 24 March 2025; Accepted: 30 May 2025

Published online: 30 September 2025

References

1. Murali, B. et al. Experimental investigations on dry sliding wear behavior of kevlar and natural Fiber-Reinforced hybrid composites through an RSM-GRA hybrid approach. *Materials* **15** (3). <https://doi.org/10.3390/ma15030749> (2022).
2. Najeeb, M. I. et al. Flexural, dynamic and Thermo-Mechanical analysis of pineapple leaf fiber/epoxy composites. *J. Nat. Fibers*. **19**, 15930–15947. <https://doi.org/10.1080/15440478.2022.2139323> (2022).
3. Liu, H. et al. Compatibilizing and toughening bamboo flour-filled HDPE composites: Mechanical properties and morphologies, *Compos Part A Appl Sci Manuf*, vol. 39, no. 12, pp. 1891–1900, Dec. (2008). <https://doi.org/10.1016/j.compositesa.2008.09.011>
4. Verma, Y. K., Singh, A. K., Paswan, M. K. & Gurumaita, P. K. Preparation and characterization of bamboo based nanocellulose by ball milling and used as a filler for preparation of nanocomposite, *Polymer (Guildf)*, vol. 308, Aug. (2024). <https://doi.org/10.1016/j.polymer.2024.127396>
5. Xu, J. et al. Mechanical properties, morphology, and creep resistance of ultra-highly filled bamboo fiber/polypropylene composites: effects of filler content and melt flow index of polypropylene. *Constr. Build. Mater.* **310** <https://doi.org/10.1016/j.conbuildmat.2021.125289> (Dec. 2021).
6. Li, L. et al. Sustainable engineering polymer composites fabricated using delignified bamboo fiber as reinforcement and walnut shell powder as filler. *Compos. B Eng.* **287** <https://doi.org/10.1016/j.compositesb.2024.111857> (Dec. 2024).
7. Prabhu, R., Mendonca, S., Bellairu, P. K., DSouza, R. C. & Bhat, T. Effect of TiO₂ filler on mechanical and tribological properties of Owen bamboo fiber reinforced epoxy composite, *World Journal of Engineering*, vol. 21, no. 4, pp. 781–792, Jun. (2024). <https://doi.org/10.1108/WJE-12-2022-0495>
8. Alfeki, M. A. & Feyissa, E. A. Water Absorption, Thermal, and Mechanical Properties of Bamboo Fiber with Chopped Glass Fiber Filler-Reinforced Polyester Composites, *Advances in Materials Science and Engineering*, vol. 2024, (2024). <https://doi.org/10.1155/2024/6262251>
9. Gouda, K., Bhowmik, S. & Das, B. Thermomechanical behavior of graphene nanoplatelets and bamboo micro filler incorporated epoxy hybrid composites. *Mater. Res. Express*. **7** (1). <https://doi.org/10.1088/2053-1591/ab67f8> (2020).
10. Gupta, A. Improvement of Physiochemical Properties of Short Bamboo Fiber-Reinforced Composites Using Ceramic Fillers, *Journal of Natural Fibers*, vol. 17, no. 11, pp. 1582–1593, Nov. (2020). <https://doi.org/10.1080/15440478.2019.1584079>
11. Ramu, S., Senthilkumar, N., Rajendran, S., Deepanraj, B. & Paramasivam, P. Thermal Conductivity and Mechanical Characterization of Bamboo Fiber and Rice Husk/MWCNT Filler Epoxy Hybrid Composite, *J Nanomater.*, vol. 2022, (2022). <https://doi.org/10.1155/2022/5343461>

12. Lee, S. Y., Kang, I. A., Park, B. S., Doh, G. H. & Park, B. D. Effects of Filler and Coupling Agent on the Properties of Bamboo Fiber-Reinforced Polypropylene Composites, *Journal of Reinforced Plastics and Composites*, vol. 28, no. 21, pp. 2589–2604, Nov. (2009). <https://doi.org/10.1177/0731684408094070>
13. Pulikkalparambil, H. et al. Effect of graphite fillers on woven bamboo fiber-reinforced epoxy hybrid composites for semistructural applications: fabrication and characterization. *Biomass Convers. Biorefin.* **14**, 17761–17777. <https://doi.org/10.1007/s13399-023-03811-y> (Aug. 2024).
14. Chatterjee, A. & Islam, M. S. Fabrication and characterization of TiO₂-epoxy nanocomposite. *Mater. Sci. Engineering: A*. **487**, 1–2. <https://doi.org/10.1016/j.msea.2007.11.052> (Jul. 2008).
15. Halász, G., Gyüre, B., János, I. M., Szabó, K. G. & Tél, T. Vortex flow generated by a magnetic stirrer. *Am. J. Phys.* **75** (12), 1092–1098. <https://doi.org/10.1119/1.2772287> (2007).
16. James, J. D. D., Mohan, R. & Vijay, R. Effect of stacking sequence on tribological properties of bamboo/jute reinforced hybrid epoxy polymer composites, in *Materials Today: Proceedings*, Elsevier Ltd, pp. 1–5. (2020). <https://doi.org/10.1016/j.matpr.2020.04.809>
17. Sathyamoorthy, G., Vijay, R. & Lenin Singaravelu, D. Development and characterization of alkali-treated and untreated Dactyloctenium aegyptium fibers based epoxy composites, in *Materials Today: Proceedings*, Elsevier Ltd, pp. 1215–1220. (2020). <https://doi.org/10.1016/j.matpr.2020.03.796>
18. Gurmu, D. N., Gebrelbanos, H. M., Tefera, C. A. & Sirahbizu, B. Experimental investigation the effect of bamboo micro filler on performance of bamboo-sisal-E-glass fiber-reinforced epoxy matrix hybrid composites. *Heliyon* **10** (22). <https://doi.org/10.1016/j.heliyon.2024.e40176> (Nov. 2024).
19. Saleh, B. et al. Optimized mechanical properties of magnesium matrix composites using RSM and ANN. *Mater. Sci. Eng. B Solid State Mater. Adv. Technol.* **290**, 116303. <https://doi.org/10.1016/j.mseb.2023.116303> (2023).
20. Saravana Kumar, M., Jayaprakash, N. & Yang, C. H. Influence of defects formation on the cracking behavior of LPBFed Cu-Cr-Zr alloy: Micro-mechanism and CRITIC-WASPAS approach. *Eng. Fail. Anal.* **160**, 108206. <https://doi.org/10.1016/j.engfailanal.2024.108206> (2024).
21. Murugesan, S. K., Natarajan, J., Yang, C. H. & Vijayananth, K. A synergistic impact of LPBF process parameters on attaining a defect-free Cu-Cr-Zr alloy parts: an analytical and experimental study. *Int. J. Adv. Manuf. Technol.* **128**, 7–8. <https://doi.org/10.1007/s00170-023-12179-2> (2023).
22. Kavimani, V., Gopal, P. M., Sumesh, K. R., Radhika, N. & Giri, J. Optimization of WEDM parameters for machining Mg-Li-RE alloy using CRITIC-COCOSO approach. *Int. J. Interact. Des. Manuf.* <https://doi.org/10.1007/s12008-024-01913-z> (May 2024).
23. Dutta, H., Gurusamy, P. & Halder, S. Enhancement of thermal and mechanical properties of polymer composite using a novel milled bamboo filler, *Polym Compos*, vol. 44, no. 12, pp. 8731–8749, Dec. (2023). <https://doi.org/10.1002/pc.27733>
24. Khan, A. et al. Extraction and characterization of natural fiber from Eleusine indica grass as reinforcement of sustainable fiber reinforced polymer composites. *J. Nat. Fibers*. **18** (11), 1742–1750. <https://doi.org/10.1080/15440478.2019.1697993> (2021).
25. Qiao, Y. et al. Phenolic-based porous composite with embedded short carbon fiber/hollow spheres for mechanical properties and thermostability. *Compos. Commun.* **55** <https://doi.org/10.1016/j.coco.2025.102305> (Apr. 2025).
26. Shahapurkar, K. et al. Parametric analysis of epoxy/crumb rubber composite by using Taguchi—GRA hybrid technique. *Polym. (Basel)*. **13**, 1–12. <https://doi.org/10.3390/polym13193441> (2021).
27. Su, Y. et al. Statistical effects of pore features on mechanical properties and fracture behaviors of heterogeneous random porous materials by phase-field modeling. *Int J Solids Struct*, vol. 264, Mar. (2023). <https://doi.org/10.1016/j.ijsolstr.2022.112098>
28. Zhu, S., Fu, Y., Wu, S. & Guo, S. Numerical simulation of void elimination in the billet during hot shape rolling processes based on the Gurson-Tvergaard-Needleman model, *AIP Adv.*, vol. 14, no. 2, Feb. (2024). <https://doi.org/10.1063/5.0186700>
29. Hashim, J., Looney, L. & Hashmi, M. S. J. The wettability of SiC particles by molten aluminium alloy. *J. Mater. Process. Technol.* **119**, 1–3. [https://doi.org/10.1016/S0924-0136\(01\)00975-X](https://doi.org/10.1016/S0924-0136(01)00975-X) (2001).
30. Saba, N. & Jawaid, M. A review on thermomechanical properties of polymers and fibers reinforced polymer composites. *Nov 25 2018 Korean Soc. Industrial Eng. Chemistry* <https://doi.org/10.1016/j.jiec.2018.06.018>
31. Tengsuthiwat, J., G. V. A. V. R. Y. G. T., Rangappa, S. M. & Siengchin, S. Characterization of novel natural cellulose fiber from Ficus Macrocarpa bark for lightweight structural composite application and its effect on chemical treatment. *Heliyon* **10** (9). <https://doi.org/10.1016/j.heliyon.2024.e30442> (May 2024).
32. Kumar, R., Kumar, K., Bhowmik, S. & Sarkhel, G. Tailoring the performance of bamboo filler reinforced epoxy composite: insights into fracture properties and fracture mechanism. *J. Polym. Res.* **26** (2). <https://doi.org/10.1007/s10965-019-1720-x> (Feb. 2019).
33. Raghunathan, V. et al. Sustainable characterization of brake pads using raw/silane-treated Mimosa pudica fibers for automobile applications, *Polym Compos*, vol. 45, no. 11, pp. 10204–10219, Aug. (2024). <https://doi.org/10.1002/pc.28467>
34. Wang, Y., Niu, Z., Han, H., Wang, J. & Lan, D. Observation of structural, mechanical, thermal and microwave dielectric properties of carbon black reinforced PA6/HDPE nanocomposites. *J. Mater. Sci.: Mater. Electron.* **34** (28). <https://doi.org/10.1007/s10854-023-11360-3> (Oct. 2023).
35. Agarwal, G., Patnaik, A. & Kumar Sharma, R. Thermo-mechanical properties of silicon carbide-filled chopped glass fiber-reinforced epoxy composites. [Online]. Available: <http://www.advancedstructeng.com/content/5/1/21>
36. Wang, Y. et al. Perforated piezoresistive film-based flexible bidirectional strain sensors for large bending deformation detection and health monitoring of glass fiber-reinforced polymers. *Compos. B Eng.* **293** <https://doi.org/10.1016/j.compositesb.2025.112111> (Mar. 2025).
37. Latha, P. S. & Rao, M. V. Investigation into effect of ceramic fillers on mechanical and tribological properties of Bamboo-Glass hybrid Fiber reinforced polymer composites. *Silicon* **10** (4), 1543–1550. <https://doi.org/10.1007/s12633-017-9637-7> (Jul. 2018).
38. Raghunathan, V., Ayyappan, V., Rangappa, S. M. & Siengchin, S. Development of fiber-reinforced polylactic acid filaments using untreated/silane-treated trichosanthes cucumerina fibers for additive manufacturing, *Journal of Elastomers and Plastics*, vol. 56, no. 3, pp. 277–292, Apr. (2024). <https://doi.org/10.1177/00952443241229186>
39. Hameed, N., Sreekumar, P. A., Valsaraj, V. S. & Thomas, S. High-performance composite from epoxy and glass fibers: Morphology, mechanical, dynamic mechanical, and thermal analysis, *Polym Compos*, vol. 30, no. 7, pp. 982–992, Jul. (2009). <https://doi.org/10.1002/pc.20644>
40. Sun, M. et al. Mechanical properties of a novel mechanical metamaterial with multi-stable and mono-stable characteristics. *Thin-Walled Struct.* **212** <https://doi.org/10.1016/j.tws.2025.113230> (Jul. 2025).

Acknowledgements

The authors extend their appreciation to the King Saud University for support through the Research Institute/Center Supporting Program (RICSP-25-2), King Saud University, Riyadh, Saudi Arabia.

Author contributions

M. Saravana Kumar, Che-Hua Yang: Conceptualization; Investigation; Visualization; Writing – Original Draft; Writing – Review & Editing, Khaja Moiduddin, Syed Hammad Mian: Conceptualization; Visualization; Analysis; Writing – Review & Editing, V. Kavimani, Adeolu A. Adediran: Data curation and validation.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.S.K. or A.A.A.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025